

Name : _____

Python Complete Guide

Q1

Operations and computational graphs of Tensorflow can run without session

- A. True
- B. False

Q2

In TensorFlow, a graph contains nodes and edges. The former represents data operations and the later represent data and its dependency relationship.

- A. True
- B. False

Q3

(Single-Answer Question) Which of the following methods CANNOT get data into Tensorflow program?

- A. Editing a file format reader
- B. Preloaded data
- C. Feeding
- D. Reading from files

Q4

(Multiple-Answer Question) What are the implementation modes of TensorFlow?

- A. Forward mode
- B. Reverse mode
- C. Distributed mode
- D. Single-Node mode

Q5

(Single-Answer Question) Which of the following is NOT a tensor attribute?

- A. Name (Logo)
- B. Shape (Dimensionality)
- C. Type (Type)
- D. Node (Node)

Q6

(Single-Answer Question) Which of the following options is NOT the session mode used by TensorFlow?

- A. Explicitly call the session to generate function
- B. Explicitly call the session to close function
- C. Through the Python context manager
- D. Multiple POST queries

Name : _____

Q7

(Single-Answer Question) Which of the following statement about the fetch operation in TensorFlow is TRUE?

- A. Only a single tensor node can be fetched once.
- B. Generally, the fetch operation is used together with `tf.placeholder()`.
- C. Only when the graph execution is triggered, you can fetch the tensor values you want.
- D. You can fetch the tensor values you want once the graph is created.

Q8

(Single - Answer Question) Which of the following files is the graph structure of Tensorflow saved in?

- A. checkpoint file
- B. index file
- C. cpkt file
- D. metafile

Q9

Data in TensorFlow is represented by tensors.

- A. True
- B. False

Q10

(Single-Answer Question) Which of the following is NOT a module in the Tensorflow library?

- A. `tf.boost`
- B. `tf.nn`
- C. `tf.layers`
- D. `tf.contrib`

Q11

Tensorflow is Google's first generation of proprietary machine learning system.

- A. True
- B. False

Q12

Which of the following options is NOT a tensor attribute?

- A. node
- B. type
- C. shape
- D. name (identification)

Q13

In Tensorflow, data is expressed in the form of tensors and computational graphs

- A. True
- B. False

Name : _____

Q14

The neural network training process will generate a lot of data. What mechanism does Tensorflow use to avoid the problem of too much input data?

- A. placeholder
- B. fetch
- C. feed
- D. client

Q15.

Which of the following environments does NOT support the installation of Tensorflow?

- A. Openstack
- B. Linux
- C. MacOS
- D. Docker

Q16

There are a lot of data generated during the training of the neural network. What mechanism does TensorFlow use to avoid excessive input data?

- A. Client
- B. feed
- C. placeholder
- D. fetch

Q17.

Which of the following data types does Tensorflow NOT support?

- A. int8
- B. float32
- C. double64
- D. bool

Q18.

Which of the following is the computational model of tensorflow?

- A. Session
- B. Tensor
- C. Computational graph
- D. Variable

Name : _____

Q19

(Multiple-Answer Question) What are the characteristics of TensorFlow? ()

- A. Open source
- B. Convenient
- C. Well-developed
- D. Flexible

Q20

(True or False) Operations and computation graph of TensorFlow do NOT necessarily run in the Session. ()

- A. TRUE
- B. FALSE

Q21

In Tensorflow, data is represented in the form of tensors and calculation charts

- A. True
- B. False

Q22

(Multiple-Answer Question) Which of the following options belong to Tensorflow concepts?

- A. Tensor
- B. Variables
- C. Placeholder
- D. Operation
- E. Conversation

Q23

(Multiple-Answer Question) What are the core concepts in Tensorflow?

- A. Gridding
- B. Calculation Chart
- C. Dot Product
- D. Tensor

Q24

(Single-Answer Question) Which of the following options is NOT the Tensorflow build process?

- A. Building a calculation chart
- B. Input tensor
- C. Generate a session
- D. Update weights

Name : _____

Q25

(Multiple-Answer Question) What are the mainstream deep learning open source tools?

- A. TensorFlow
- B. Caffe
- C. Torch
- D. Theano
- E. Scikit-learn

Q26

(Multiple-Answer Question) Which of the following are the characteristics of TensorFlow?

- A. Open source
- B. Convenience
- C. C Mature
- D. Flexible

Q27

TensorFlow supports multi-TPU cluster computing

- A. True
- B. False

Q28

Tensorflow framework does NOT support Windows systems

- A. True
- B. False

Q29

(Single-Answer Question) Which of the following is NOT part of the deep learning development framework?

- A. CNTK
- B. Keras
- C. BAFA
- D. MXNet

Q30

(Multiple-Answer Question) Which of the following are included in the application scenarios of Tensorflow?

- A. Speech recognition
- B. Face recognition
- C. Image style changes
- D. Autopilot

Name : _____

Q31

(Single-Answer Question) Which of the following is NOT a way for the TensorFlow program to read data?

- A. Preload data
- B. Feeding data
- C. Read from file
- D. Write a file format reader

Q32

(Single-Answer Question) Which of the following is the computing model of TensorFlow?

- A. Calculation chart
- B. Tensor
- C. Conversation
- D. Variable

Q33

TensorFlow ONLY supports Python and R language

- A. True
- B. False

Q34

(Multiple-Answer Question) What are algorithms supported by TensorFlow?

- A. CNN – Convolutional Neural Network
- B. ZNN
- C. RNN – Recurrent Neural Network
- D. HUT

Q35

(Single-Answer Question) Which method is NOT supported in TensorFlow to define variables?

- A. null
- B. Random number
- C. Constant
- D. Calculated from initial values of other variables

Q36

(Single-Answer Question) Tensorflow is the second generation of artificial intelligence learning system developed by:

- A. Distbelief
- B. Paleyfunction
- C. ConvexOne
- D. Infinity

Q1

Name : _____

The print function in Python 3 is enclosed in parentheses ().

A. True

B. False

Q2

Lists in Python can be delimited by square brackets ([]), with the first element at index 0.

A. True

B. False

Q3

Python tuples are enclosed in parenthesis (()) and the internal elements are separated by semicolons(;

A. True

B. False

Q4

Continue statements in Python are used to skip the rest of the statements in the body of loop for the current iteration and jump to the beginning of the loop for next iteration

A. True

B. False

Q5

time.time() function can measure the number of seconds since the epoch as a floating point value.

A. True

B. False

Q6

Python author designed the restrictive syntax to filter out bad programming (like failing to add an indentation on the right of the next line of the if statement)

A. True

B. False

Q7

Name : _____

(Single-Answer Question) Which of the following lines properly starts a parameter less function definition?

- A. function func():
- B. def func:
- C. fun function():
- D. def func():

Q8

Functions are organised code segments which CANNOT be REUSED to implement single or related operations.

- A. True
- B. False

Q9

(Single-Answer Question) Which of the following is not a Python re regular expression module function?

- A. Match
- B. Compile
- C. Update
- D. Search

Q10

(Single-Answer Question) Knowing that a function named myFunc() resides in a module named myMod, choose the proper way to import it:

- A. import myFunc() from myMod
- B. import myFunc()
- C. from myMod import myFunc()
- D. from myFunc() import myMod

Q11

(Single-Answer Question) Which of the following Python identifier is INVALID?

Name : _____

- A. 1var
- B. var_1
- C. Var_1
- D. _var1

Q12

(Single-Answer Question) Which of the following is unchangeable in Python?

- A. List
- B. Tuple
- C. Array
- D. Set

Q13

(Multiple-Answer Question) Which of the following statements are TRUE?

- A. Python is often nicknamed “glue language” because it can easily glue together various modules developed in other languages.
- B. It was created by Guido van Rossum, the Dutchman, in 1989, and it was first released in 1991.
- C. Python is a kind of free and open-source software.
- D. The Python syntax is simple and clear, characterized by using blank characters for indentation.

Q14

(Multi-Answer Question) Python is an object-oriented programming language. Which of the following are Python objects?

- A. Function
- B. Module
- C. Number
- D. Character string

Name : _____

Q15

(Multi-Answer Question) Which of the following statements about Python index are true?

- A. Default index from right to left is numbered from -1.
- B. Default index from left to right is numbered from 0.
- C. Default index from right to left is numbered from 0.
- D. Default index from left to right is numbered from 1.

Q16

(Single-Answer Question) One of the following variables' names is illegal – which one?

- A. true
- B. tTRUE
- C. True
- D. TRUE

Q17

(Single-Answer Question) An operator that is able to check whether two values are equal is coded as

- A. =
- B. ==
- C. ===
- D. !=

Q18

(Single-Answer Question) Which of the following is not a Python operator?

- A. Arithmetic operator
- B. Inference operator
- C. Logical operator
- D. Comparison operator

Name : _____

Q19

(Single-Answer Question) How many hashes will the following code snippet send to the console?

```
i = 0
while i <= 3 :
    i += 2
    print("#")
```

- A. two
- B. zero
- C. one
- D. three

Q20

(Single-Answer Question) Which of the following statements about Python regular expression is NOT true? ()

- A. re regular expression module was added in Python 1.5
- B. re module grants all regular expression functions to Python.
- C. re regular expression is a powerful tool to process strings
- D. re regular expression can only process character string data, not numerical data.

Q21

(Single-Answer Question) According to Python Enhancement Proposal (PEP) 8, how many spaces should the first line be indented by in Python? ()

- A. 1
- B. 2
- C. 4
- D. 8

Name : _____

Q22

(Multi-Answer Question) What are the numeric types of Python?

- A. int (integer type)
- B. long (long integer type)
- C. float (floating point type)
- D. complex (complex number type)

Q23

(Single-Answer Question) When a file is opened in read mode, it ():

- A. must exist (an exception will be raised otherwise)
- B. cannot exist (it has to be created every time)
- C. it will be deleted, if it exists
- D. it doesn't have to exist (it will be created if not present)

Q24

(Single-Answer Question) If you want to open a text file in append mode, you would use the following mode string:

- A. t+a
- B. at
- C. a+t
- D. at+

Q25

(Single-Answer Question) Assuming that the open() invocation has gone successfully, the following code snippet will:

```
for x in open('file','rt'):

    print(x)
```

- A. read the file line by line
- B. read the file character by character
- C. cause an exception error
- D. read the whole file at once.

Name : _____

Q26

(Single-Answer Question) What is WRONG description of the Python module?

- A. The Python module is a Python file that ends with .py and contains Python object definitions and Python statements.
- B. The Python module allows you to logically organize your Python code snippets.
- C. Python modules can define functions, classes, and variables, but the module does NOT contain executable code.
- D. Assigning relevant code to a module can make your code better and easier to understand.

Q27

The commonly used functions for mathematical operations in Python are basically in the math module and the cmath module.

- A. True
- B. False

Q28

(Single-Answer Question) What is WRONG with the introduction of Python module time?

- A. from time
- B. import time
- C. from time import*
- D. import time as t

Q29

(Multi-Answer Question) What are the advantages of Python?

- A. free
- B. simple
- C. high-level language
- D. rich library

Name : _____

Q30

(Multi-Answer Question) What are the Python language data types?

- A. list
- B. string
- C. tuple
- D. number
- E. dictionary

Q31

(Single-Answer Question) What is NOT the difference between Python 2 and Python 3?

- A. import
- B. print
- C. unicode
- D. xrange

Q32

(Multi-Answer Question) What operating systems does Python support?

- A. linux
- B. DOS
- C. Windows
- D. MacOS

Q33

Print in Python 3 must be used with ()

- A. True
- B. False

Name : _____

Q34

The components of Python identifier include: numbers, letters, and underscores

- A. True
- B. False

Q35

(Multi-Answer Question) The Python script execution model includes interactive mode and script mode

- A. True
- B. False
- C. 0
- D. Null

Q36

Python tuples are identified by “()” and internal elements are separated by “:”

- A. True
- B. False

Q37

(Multi-Answer Question) What are the advantages of Python?

- A. Beautiful
- B. Expensive
- C. Explicit
- D. Simple

Name : _____

Q38

(Single-Answer Question) What is the output of the following snippet if the user enters two lines containing 3 and 6 respectively?

```
i = input()
j = input()
Print (i + j)
```

- A. 9
- B. 36
- C. 6
- D. 3

Q39

(Single-Answer Question) The print() function can output values of:

- A. Any number of arguments (excluding zero)
- B. Just one argument
- C. Any number of arguments (including zero)
- D. Not more than five arguments

Q40

(Single-Answer Question) What is the output of the following snippet if the user enters two lines containing 2 and 4 respectively?

```
i = input()
j = int(input())
print(i * j)
```

- A. 24
- B. 8
- C. 2222
- D. 44

Name : _____

Q41

(Single-Answer Question) What is the output of the following snippet if the user enters two lines containing 3 and 5 respectively?

```
i = int(input())  
j = int(input())  
print(i + j)
```

- A. 5
- B. 3
- C. 35
- D. 8

Q42

(Single-Answer Question) Python regular expressions are a special sequence of characters that makes it easy to check if a string matches pattern.

- A. True
- B. False

Q43

(Single-Answer Question) Python language can use multiple statements on the same line, separated by commas.

- A. True
- B. False

Q44

(Multi-Answer Question) Python is a fully object-oriented language. Which of the following options belong to the Python object?

- A. Function
- B. Module
- C. Number
- D. Character string

Name : _____

Q45

(Multi-Answer Question) What quotation marks can the Python language use?

- A. Single quotes
- B. Double quotes
- C. Three quotes
- D. Four quotes

Q46

(Single-Answer Question) The Python list can be identified by "[]", and the default index of the first element from left to right is 1.

- A. True
- B. False

Q47

(Single-Answer Question) Which of the following is NOT the difference between Python 2 and Python 3?

- A. print
- B. unicode
- C. import
- D. Xrange

Q48

(Single-Answer Question) The commonly used functions for mathematical operations in Python are basically in the math module and the cmath module.

- A. True
- B. False

Q49

(Single-Answer Question) Python tuples are identified by "()" and internal elements are separated by ;

- A. True
- B. False

Name : _____

Q50

(Multi-Answer Question) Which of the following are the characteristics of the Python language?

- A. Explanatory
- B. Process oriented
- C. Object-oriented
- D. Dynamic data type

Q51

(Single-Answer Question) The timestamps in the Python language are represented by how long in seconds elapsed from midnight (epoch) on January 1, 1970

- A. True
- B. False

Q52

(Single-Answer Question) What are the results returned by the if conditional statements in the Python language?

- A. True
- B. 0
- C. False
- D. Null

Q53

(Single-Answer Question) The Python language can use the “#” at the beginning of a single line of code for code comments

- A. True
- B. False

Name : _____

Q54

(Single-Answer Question) Python authors deliberately design very restrictive grammars that make bad programming habits (such as the next line of if statement not indented to the right) cannot be compiled.

- A. True
- B. False

Q55

(Single-Answer Question) What are the steps that are NOT part of the operation of a Python file object?

- A. open
- B. delete
- C. read
- D. write
- E. close

Q56

(Single-Answer Question) The *for* loop statement in the Python language can iterate through the items in any sequence.

- A. True
- B. False

Q57

(Multi-Answer Question) About Python tuple which is CORRECT description?

- A. The difference between a list and a tuple is that the elements of the tuple CANNOT be modified.
- B. The list uses square brackets, and the tuple uses parentheses.
- C. Tuple creation is very simple, just add elements in parentheses and separate them with comma.
- D. When a tuple contains only one element, you need to add a comma after the element.

Name : _____

Q58

(Multi-Answer Question) Which is the correct description of the Python creation function?

- A. The function created starts with the def keyword followed by
- B. delete
- C. read
- D. write
- E. close

Q59

(Multi-Answer Question) What are the characteristics of Python code.

- A. Legibility
- B. Simplicity
- C. Rapidity
- D. Scalability

Q60

(Single-Answer Question) The constituent elements of the Python identifier include numbers, letters and underscores.

- A. True
- B. False

Q61

(Single-Answer Question) Which description is WRONG about the lambda function of the Python language.

- A. Lambda is just an expression, and the function body is much simpler than def
- B. The body of a lambda can be an expression or a block of code.
- C. The lambda function can access the parameters in the global namespace.
- D. The lambda function accepts only ONE parameter value.

Name : _____

Q61

(Single-Answer Question) The time interval in the Python language is a floating-point fraction in seconds.

- A. True
- B. False

Q62

(Multi-Answer Question) Which of the following descriptions about Python program are correct?

- A. The program consists of modules
- B. Module contains statements
- C. Statement contains an expression
- D. Expressions create and process objects

Q63

(Multi-Answer Question) What are the ADVANTAGES of Python?

- A. simple
- B. free
- C. high-level language
- D. rich library

Q64

(Single-Answer Question) Python script execution mode includes interactive mode and script mode.

- A. True
- B. False

Q65

(Multi-Answer Question) Which of the following features does Python support?

- A. inheritance
- B. heavy load
- C. derived
- D. multiple inheritance

Name : _____

Q66

(Single-Answer Question) Which of the following description about Python regular expression is NOT correct?

- A. Python has added the re regular expression module since version 1.5
- B. re regular expression module gives the Python language all the regular expression functionality.
- C. re regular expressions are powerful tools for working with strings.
- D. re regular expressions can only process string data, NOT numeric data.

Q67

(Multi-Answer Question) What are the application scenarios for the break statement in the Python language?

- A. Any Python statement
- B. *while* loop statement
- C. *for* loop statement
- D. Nested loop statement

Q68

(Single-Answer Question) Python dictionary is identified by {} and the internal data consists of the key and its corresponding value.

- A. True
- B. False

Q69

(Single-Answer Question) “print” in Python 3 must be used with “Queue”.

- A. True
- B. False

Q70

(Single-Answer Question) Which of the following is NOT a Python operator?

- A. Arithmetic operator
- B. Reasoning operator
- C. Logical operator
- D. Comparison operator

Name : _____

Deep Learning and Huawei EI

ASSIGNMENT ACTIVITY

Q1

Huawei believes that the value of AI is reflected in addressing enterprise business problems. Regarding the development of AI, guided by industry solutions, Huawei focuses on basic technologies and the enabling platform and works with industry leaders to develop industry solutions

A. True

B. False

Q2

Deep learning is a class of machine learning algorithms that uses neural networks to extract features from the raw input.

A. True

B. False

Q3

Convolutional neural networks (CNN) can extract features from images automatically

A. True

B. False

Q4

Deep learning algorithms can be both supervised and unsupervised

A. True

B. False

Q5

The perceptron is equivalent to a classifier and it could perform multi-task classification.

A. True

B. False

Name : _____

Q6

Recurrent neural networks are more suitable for image recognition problems.

- A. True
- B. False

Q7

(Single-Answer Question) Which is NOT an automatic hyperparameter search algorithm?

- A. Grid search
- B. Random search
- C. Random gradient descent
- D. Model-based hyperparameter optimization

Q8

(Single-Answer Question) Which of the following activation functions CANNOT effectively solve the vanishing gradient problem?

- A. Leaky ReLU
- B. Tanh
- C. ReLU
- D. SoftPlus

Q9

(Single-Answer Question) In the input of a softmax function is [1,2,4,2,1], which of the following option may be the output?

- A. [0.04, 0.20, 0.75, 0.20, 0.04]
- B. [0.04, 0.10, 0.72, 0.10, 0.04]
- C. [10]
- D. [3]

Q10

(Single-Answer Question) What is the subject of artificial intelligence?

- A. Mathematics and Physiology
- B. Psychology and Physiology
- C. Linguistics

Name : _____

D. Comprehensive interdisciplinary and marginal disciplines

Q11

(Single-Answer Question) Which is NOT an optimization method in deep learning?

- A. Stochastic gradient descent
- B. Principle component analysis (PCA)
- C. Backpropagation
- D. Momentum

Q12

(Single-Answer Question) Which of the following is NOT a deep learning development framework?

- A. CNTK
- B. BAFA
- C. MXNet
- D. Keras

Q13

(Single-Answer Question) Decision tree is one of the most popular machine learning algorithms and can be used for both classification and regression problems. How do we choose the best attribute in ID3?

- A. The attribute with the highest information gain
- B. The attribute with the lowest information gain
- C. The attribute with the lowest Gini index
- D. The attribute with the highest Gini index

Q14

(Single-Answer Question) Which of the following is NOT a supervised learning algorithm?

- A. Principle component analysis (pca)
- B. Logistic regression
- C. Support vector machine (SVM)
- D. Decision tree

Name : _____

Q15

(Single-Answer Question) What is the purpose of adding a pooling layer when creating a neural network?

- A. Reduce the size of feature maps
- B. Extracting image feature
- C. Preventing overfitting
- D. Function as a classifier

Q16

(Single-Answer Question) Which of the following is used as the gate function in LSTM?

- A. Relu function
- B. Softplus function?
- C. Sigmoid function
- D. Tanh function

Q17

(Single-Answer Question) Which of the following can avoid overfitting during model training?

- A. Activation functions
- B. Adding the number of convolutional layers
- C. Early stopping
- D. Using momentum optimizer

Q18

(Single-Answer Question) Which of the following statement about convolutional neural networks is NOT true?

- A. Pooling layers in CNN are implemented to reduce the size of feature maps.
- B. A convolutional layer contains a set of filters whose parameters need to be learned.
- C. Max pooling is the most commonly used pooling operation.

Name : _____

D. A pooling layer contains a set of filters whose parameters need to be learned.

Q19

(Single-Answer Question) Which of the following type of RNN can be used for machine translation?

- A. Many to one
- B. Many to many
- C. One to many
- D. One to one

Q20

(Single-Answer Question) Which of the following school does reinforcement learning belong to?

- A. Behaviourism
- B. Connectionism
- C. Symbolism
- D. None of the above

Q21

(Multi-Answer Question) For object detection problem, which of the following neural network model should be considered first?

- A. CNN
- B. RNN
- C. GAN
- D. GNN

Q22

(Multiple-Answer Question) What are the research areas in AI?

- A. Natural language processing (NLP)
- B. Computer vision
- C. Speech recognition
- D. Machine learning

Name : _____

Q23

(Multiple-Answer Question) Which of the following cloud services are provided by visual recognition of Huawei Cloud EI?

- A. Optical Character Recognition (OCR)
- B. Image Recognition
- C. Content Moderation
- D. Image processing

Q24

(Multiple-Answer Question) Which of the neural network architectures can implement weight sharing?

- A. Convolutional neural network (CNN)
- B. Recurrent neural network (RNN)
- C. Fully connected neural network (FCN)
- D. All of the above

Q25

(Multiple-Answer Question) What are the elements of AI? ()

- A. Algorithms
- B. Computing performance
- C. Data
- D. Scenarios

Q26

(Multiple-Answer Question) Training error causes underfitting by reducing model accuracy. How can the model be a better fit? ()

- A. Increasing data volume
- B. Creating feature engineering
- C. Reducing regularization parameters
- D. Adding features

Name : _____

Q27

(Single-Answer Question) Which of the following statements about gradient descent is NOT true? ()

- A. Stochastic gradient descent is a common form of gradient descent.
- B. Gradient descent includes stochastic gradient descent and batch gradient descent.
- C. The gradient descent algorithm is fast and reliable.
- D. Stochastic gradient descent is one of the commonly used optimization algorithms in Deep Learning.

Q28

(Single-Answer Question) Which of the following is NOT involved in the three types of services for Huawei Cloud EI to construct enterprise intelligent services? ()

- A. Basic platform services
- B. General domain services
- C. Industry services
- D. Integration services

Q29

(Single-Answer Question) There are a lot of data generated during the training of the neural network. What mechanism does TensorFlow use to avoid excessive input data?

- A. Client
- B. feed
- C. placeholder
- D. fetch

Q30

(Multiple-Answer Question) There are many commercial applications for machine learning services. What are the main business scenarios covered?

- A. Financial product recommendation

Name : _____

- B. Predictive maintenance
- C. Telecom customer retention
- D. Retailer grouping

Q31

(Multiple-Answer Question) What of the following does belong to convolutional neural network (CNN)?

- A. VGGNet
- B. ResNet
- C. AlexNet
- D. GoogleNet

Q32

(Single-Answer Question) What is WRONG description of backpropagation?

- A. The learning process of the backpropagation algorithm consists of a forward propagation process and a backpropagation process.
- B. The backpropagation algorithm is a learning algorithm suitable for multi-layer neural networks, which is based on the gradient descent method.
- C. The backpropagation phase sends training inputs to the network to obtain an stimuli response.
- D. The backpropagation algorithm is mainly repeated by two loops (excitation propagation, weight update) until the response of the network to the input reaches the predetermined target range.

Q33

(Single-Answer Question) Which of the following options is NOT a reason for traditional machine learning algorithms to promote the development of deep learning?

- A. Dimensional disaster
- B. Local invariance and smooth regularization
- C. Manifold learning
- D. Feature Engineering

Name : _____

Q34

(Single-Answer Question) What does NOT belong to supervised learning?

- A. Logistic regression
- B. Support vector machine
- C. Decision tree
- D. Principal component analysis

Q35

(Multiple-Answer Question) The following statements about overflow and underflow in numerical calculations are correct?

- A. Underflow occurs when the numbers close to zero are rounded to zero
- B. Overflow occurs when a large number of levels are approximate to positive infinity or negative
- C. Underflow occurs when a large number of levels are approximate to positive infinity or negative
- D. An overflow occurs when the numbers close to zero are rounded to zero

Q36

The autoencoder is an unsupervised learning algorithm

- A. True
- B. False

Q37

(Multiple-Answer Question) What are the service categories included in Huawei Cloud EI Enterprise Intelligence? (Multiple Choice)

- A. EI visual cognition
- B. EI online games
- C. EI speech semantics
- D. EI industry scene

Name : _____

Q38

Generating Artificial Network is a Deep Learning model. Recently, it is one of the most promising methods for unsupervised learning in complex distribution. ()

- A. True
- B. False

Q39

Convolutional neural networks are commonly used to process images ()

- A. True
- B. False

Q40

When the inputs are the same, the Feedforward neural networks (FNN and CNN) will always have the same outputs. ()

- A. True
- B. False

Q41

LSTM model can alleviate the vanishing gradient problem in RNN. ()

- A. True
- B. False

Q42

The gradient explosion problem can be solved with simple gradient clipping method. ()

- A. True
- B. False

Name : _____

Q43

GAN can use unsupervised learning to generate images ()

- A. True
- B. False

Q44

(Multiple-Answer Question) Which of the method can alleviate overfitting problem in Deep Learning?

- A. L1 regularization
- B. L2 regularization
- C. Dataset expansion
- D. Dropout

Q45

(Multiple-Answer Question) According to constraints, what are the types of the optimization problem?

- A. Equality constraint
- B. No constraint
- C. Inequality constraint
- D. Semi-constraint

Q46

Machine Learning Service is a data mining and analytics platform service that helps users quickly discover data patterns and build predictive models through machine learning techniques and deploy them as predictive analytics solutions

- A. True
- B. False

Q47

(Single-Answer Question) In a neural network, knowing the weight and deviations of each neuron is the most important step if you know the exact weights and deviations of neurons in some way, you can approximate any function. What is the best way to achieve this?

- A. Random assignment, pray that they are correct

Name : _____

- B. Search for a combination of weight and deviation until the best value is obtained.
- C. Assign an initial value to iteratively update weight by checking the difference between the best value and the initial.
- D. The above is NOT correct.

Q48

(Multiple-Answer Question) What are the regularizations in deep learning?

- A. L1 Norm, L2 Norm
- B. Data enhancement
- C. Integration method
- D. Dropout

Q49

(Single-Answer Question) Which of the following is NOT included in the Recurrent Neural Network usage scenario?

- A. Machine translation
- B. Speech recognition
- C. Image style migration
- D. Text generation

Q50

Convolutional neural networks are more suitable for dealing with speech recognition.

- A. True
- B. False

Q51

(Single-Answer Question) Which model is NOT cyclic neural network?

- A. RNN
- B. LSTM
- C. GRU
- D. GBDT

Name : _____

Q52

Convolutional neural networks are more suitable for dealing with image recognition problems than cyclic neural networks.

A. True

B. False

Q53

(Multi-Answer Question) What are the commonly used loss functions?

A. Mean variance

B. Sigmoid cross entropy

C. Softmax cross entropy

D. Sparse cross entropy

E. Weighted Sigmoid cross entropy

Q54

(Single-Answer Question) Which of the following is TRUE about unsupervised learning?

A. Unsupervised algorithm only processes “features” and does NOT process tags.

B. Dimensionality reduction algorithm is NOT unsupervised learning.

C. K-means algorithm and SVM algorithm belong to unsupervised learning.

D. None of the above

Q55

(Single-Answer Question) Which of the following is NOT Tensorflow build process?

A. Building calculation chart.

B. Input tensor

C. Generate session

D. Update weights

Q56

Cyclic neural networks are more suitable for dealing with image recognition problems.

Name : _____

- A. True
- B. False

Q57

GAN is a deep learning model and one of the most promising methods for unsupervised learning in complex distribution in recent years.

- A. True
- B. False

Q58

(Multi-Answer Question) What are the commonly used activation functions?

- A. Sigmoid Building calculation chart.
- B. Tanh
- C. Relu
- D. Danish

Q59

(Single-Answer Question) What is the INCORRECT relationship between neural networks and deep learning?

- A. The concept of deep learning stems from the study of artificial neural networks.
- B. A neural network algorithm with multiple hidden layers is a deep learning algorithm
- C. Single layer neural networks are also a type of deep learning
- D. Convolutional neural networks are a type of deep learning

Q60

Deep learning is a branch of machine learning

- A. True
- B. False

Q61

Name : _____

Deep learning makes it easy to derive simple mathematical functions from a large amount of high-dimensional data to describe complex decision interfaces.

- A. True
- B. False

Q62

(Single-Answer Question) Which description is WRONG about hyperparameters?

- A. Hyperparameters are parameters that set values before the algorithm begins learning
- B. Most machine learning algorithms have hyper parameters
- C. Hyper parameters CANNOT be modified
- D. The value of the hyper parameter is NOT learned by the algorithm itself.

Q63

(Single-Answer Question) Which of the following is NOT a deep learning algorithm?

- A. Self-encoder
- B. Convolutional Neural Networks
- C. Support Vector Machine
- D. Recurrent Neural Network

Q64

(Single-Answer Question) Which of the following is NOT a deep learning algorithm?

- A. Symbolism
- B. Connectionism
- C. Behaviourism
- D. None of the above

Q65

(Multi-Answer Question) Reducing the gap between the training error will result in over-fitting. How to prevent overfitting?

- A. Cross validation

Name : _____

- B. Integration method
- C. Increase regularization
- D. Feature Engineering

Q66

(Single-Answer Question) In order for a machine to be intelligent, it must be knowledgeable. Therefore, there is a research field in artificial intelligence, which mainly studies how computers automatically acquire knowledge and skills to achieve self-improvement. What is the branch of this research called?

- A. Expert system
- B. Machine learning
- C. Neural network
- D. Natural Language Processing

Q67

(Single-Answer Question) Which of the following descriptions about Recurrent Neural Network is correct?

- A. Can be used to process sequence data
- B. CANNOT process valuable length sequence data
- C. Unlike convolutional neural networks, parameters of RNN CANNOT be shared
- D. The units above the hidden layer are NOT associated with each other

Q68

(Multi-Answer Question) Training error will reduce the accuracy of the model and produce under-fitting. How to improve the model fit?

- A. Increase the amount of data
- B. Feature Engineering
- C. Reduce regularization parameters
- D. Add features

Q69

Name : _____

(Single-Answer Question) Which of the following does NOT belong to long-term memory LSTM (Long Short-Term Memory) network architecture?

- A. Memory door
- B. Forget the door
- C. Input door
- D. Output door

Q70

(Single-Answer Question) Which of the following functions can numerically stabilize overflow and underflow?

- A. Softminus function
- B. Softplus function
- C. Softmax function
- D. Softmin function

Q71

(Single-Answer Question) Which of the following does the convolutional neural network structure NOT include?

- A. Convolutional layer
- B. Pooling layer
- C. Loop layer
- D. Fully connected layer

Q72

(Single-Answer Question) Which of the following description is WRONG about the image recognition cloud service?

- A. The service identifies
- B. Pooling layer
- C. Loop layer
- D. Fully connected layer

Q73

Name : _____

(Multi-Answer Question) What are the commonly used gradient descent optimization functions?

- A. Random gradient descent
- B. Adadelta
- C. Adagrad
- D. Momentum
- E. RMSProp

Q74

(Single-Answer Question) Which of the following descriptions of the depth feedforward is CORRECT?

- A. Deep feed forward is a type of neural network.
- B. The deep feed forward network has only ONE hidden layer
- C. The unit on the hidden layer of the deep feedforward network will have countless
- D. Deep feedforward network is used to deal with linear problems

Q75

(Multi-Answer Question) In Huawei Cloud EI Enterprise Intelligence, which basic platform services are included?

- A. Machine learning
- B. Deep learning
- C. Graph engine
- D. Batch processing

Q76

(Multi-Answer Question) Which of the following are solutions for the Huawei Cloud EI industry scenario?

- A. Intelligence Logistics
- B. Intelligent Water
- C. Intelligent Transportation
- D. Intelligent Finance
- E. Intelligent Manufacturing

Name : _____

MATHS BASICS ASSIGNMENT ACTIVITY

Q1

The determinant of $N \times N$ matrix is a scalar.

- A. True
- B. False

Q2

Matrix A and Matrix B can only be added if they have equal number of rows and columns

- A. True
- B. False

Q3

Matrix A can be decomposed into a product of THREE matrices using singular value decomposition.

- A. True
- B. False

Q4

(Single-Answer Question) Which of the following about normal distribution is NOT TRUE?

- A. The larger the standard deviation, the steeper the normal distribution curve.
- B. In natural phenomena and social phenomena, a large number of random variables obey or approximate a normal distribution.
- C. The normal distribution takes the maximum value at the mean.
- D. The larger the standard deviation, the slower the normal distribution curve.

Q5

(Single-Answer Question) Which of the following about matrix is NOT TRUE?

- A. Multiplying by the identity matrix does NOT change anything.
- B. The diagonal matrix does NOT have an inverse.
- C. The transposed matrix of an orthogonal matrix is its inverse.
- D. The transposed matrix of a symmetric matrix A is A.

Name : _____

Q6

(Single-Answer Question) Information theory is a branch of applied mathematics. It studies the quantification of information in a signal. Who was the first to put forward the concept of information entropy?

- A. Marvin Lee Minsky
- B. Alan Mathison Turing
- C. Claude Elwood Shannon
- D. Herbert A. Simon

Q7

The trace operation returns the sum of the diagonal elements of the matrix. So the trace of matrix A is the same as that of its transposed matrix. ()

- A. TRUE
- B. FALSE

Q8

(Single-Answer Question) Which of the following is NOT the main idea of linear algebra? ()

- A. Probability theory
- B. Linear transformation
- C. Matrix theory
- D. Vector space

Q9

(Single-Answer Question) Matrix A is a 3×2 matrix, B is a 2×3 matrix, and C is a 3×3 matrix. Which of the following operations is valid? ()

- A. BC
- B. AC
- C. A+B
- D. AB-BC

Name : _____

Q10

Principal Component Analysis (PCA) is a statistical method. It uses orthogonal transformation to convert a set of possibly correlated variables into a set of values of linearly dependent variables called principal components. ()

- A. True
- B. False

Q11

Correlation coefficient, also called the linear correlation coefficient, measures a linear relationship between two variables. Its value is a real number greater than 0. ()

- A. True
- B. False

Q12

(Single-Answer Question) Suppose X and Y are random variables and C is a constant, which of the following about mathematical expectation properties is NOT true? ()

- A. $E(C)=C$
- B. $E(X+Y)=E(X)+E(Y)$
- C. $E(CX)=CE(X)$
- D. $E(XY) = E(X)E(Y)$

Q13

With regard to the relationship between frequency and probability, which of the following statements is true?

- A. Frequency is equal to probability
- B. Frequency is smaller than probability
- C. Frequency is greater than probability
- D. Frequency stabilizes as a value close to probability as the number of repeated experiments increases

Name : _____

Q14

(Single-Answer Question) Which of the following about the description of expectations and variances is INCORRECT?

- A. Expectation reflects the average level of random variable values.
- B. The variance reflects the degree of deviation between the random variable and its mathematical expectation.
- C. Expectation and variance are both nominal characteristics of random variables.
- D. The greater the expectation, the smaller the variance.

Q15

(Single-Answer Question) Which of the following conditions is NOT a condition that n -fold Bernoulli trials need to meet?

- A. Each test was repeated under the same condition.
- B. There are only TWO possible outcomes for each trial
- C. Each test itself is obeying normal distribution.
- D. The results of each trial are independent of each other.

Q16

(Single-Answer Question) What are the conditions for m row n column matrix A and p row q column matrix B.

- A. $m=p, n=q$
- B. $n=p$
- C. $m=n$
- D. $p=q$

Q17

(Multiple – Select Answer) Which of the following conditions do the randomized trials need to meet?

- A. It can be repeated under the same conditions
- B. There may be more than one possible outcome for each trial and all possible outcomes of the trial can be clarified in advance.
- C. All results of the test CANNOT be clarified in advance
- D. It is not possible to determine which result will appear before conducting a test.

Name : _____

Q18

(Multiple-Answer Question) The matrix produced by the matrix A rows and columns is called the transpose of A. What are the correct properties of matrix transposition?

- A. $(A^T)^T = A$
- B. $(A + B)^T = A^T + B^T$
- C. $(A^T)^T = A$
- D. $(AB)^T = A^T + B^T$

Q19

(Multiple – Select Answer) Numerical calculations refer to the method and process of effectively using the digital computer to solve the approximate problem of mathematical problems, and the discipline consisting of related theories. Which of the following processes are involved in solving actual problems with a computer?

- A. Know the actual problem
- B. Mathematical model
- C. Numerical calculation method
- D. Programming
- E. Computer calculation results

Q20

(Single-Answer Question) The following are the description of the distribution function, distribution law, and density function of random variables, which is INCORRECT?

- A. Distribution law can only describe the value rule of discrete random variables
- B. The density function can only describe the value rule of continuous random variables
- C. Discrete random variables have no distribution function
- D. Distribution function describes the value rule of random variables.

Name : _____

Q21

(Single-Answer Question) According to the definition of information entropy, what is the bit entropy of throwing a uniform coin?

- A. 0
- B. 0.5
- C. 1
- D. -1

Q22

(Single-Answer Question) Suppose X and Y are random variables and C is a constant, which of the following about mathematical expectation properties is NOT true?

- A. $D(C)=0$
- B. $D(X+Y)=D(X)+D(Y)$
- C. $D(CX)=C \cdot C \cdot D(X)$
- D. $D(XY) = D(X)D(Y)$

Q23

A vector is a number.

- A. True
- B. False

Q24

(Single-Answer Question) About Bayesian formula $P(W|X) = P(X|W)/P(X)$. What is the correct description?

- A. $P(W|X)$ is a prior probability
- B. $P(X|W)$ is a conditional probability
- C. $P(W)$
- D. $P(X)$

Name : _____

Q25

(Single-Answer Question) The number of rows and columns of the three matrices A, B, and C is 3 rows and 2 columns, 2 rows and 3 columns, 3 rows and 3 columns, which of the following operations is meaningful?

- A. AC
- B. BC
- C. AB
- D. A+B

Q26

(Single-Answer Question) As shown in the figure below, what is the value of the determinant A?

$\begin{vmatrix} 1 & 2 & 4 & 5 & 7 & 8 & 3 & 6 & 9 \end{vmatrix} A =$

- A. 24
- B. 18
- C. -24
- D. 0

Q27

(Single-Answer Question) Which of the following statement about the matrix is INCORRECT?

- A. The arbitrary matrix is multiplied by the unit matrix and will not change
- B. The transpose of the symmetric matrix A is also A itself
- C. The transpose of the orthogonal matrix A is equal to the inverse of A
- D. There is no inverse matrix for the diagonal matrix

Q28

The number or matrix is stored in the tensor.

- A. True
- B. False

Q29

(Single-Answer Question) Which of the following description of the number of rows and columns of the matrix is CORRECT?

- A. The number of rows is greater than the number of columns
- B. The number of rows is equal to the number of columns
- C. The number of rows is less than the number of columns
- D. The number of rows has no relationship with the number of columns

Name : _____

Q30

(Single-Answer Question) Which of the following description about the relationship between the Poisson distribution and the binomial distribution is CORRECT?

- A. The mathematical model of the Poisson distribution and the binomial distribution are both Lagrangian models
- B. Poisson distribution is an approximation of the binomial distribution when n is very large and p is small
- C. Poisson distribution has no relationship with binomial distribution
- D. Poisson distribution can replace binomial distribution

Q31

(Single-Answer Question) Vector group $a_1 = [1,1,1]$, $a_2 = [0,2,5]$, $a_3 = [1,3,6]$. Which of the following options is CORRECT?

- A. Linear correlation
- B. Linear independence
- C. $a_1 + a_2 + a_3 = 0$
- D. $2a_1 + a_2 + a_3 = 0$

1. Operations and computation graphs of TensorFlow can be run without a session?

False

2. A matrix can be decomposed into a product of three matrices using singular value decomposition.

True

3. Deep learning algorithms can be both supervised and unsupervised.

True

4. The print function in Python is enclosed in parenthesis `()`.

True

Name : _____

5. In TensorFlow, a graph contains nodes and edges. The former represent data operations and the latter represent data and its dependency relationships.

True

6. The perception is equivalent to a classifier and it could perform multi-task classification.

False

7. Data in TensorFlow is represented by tensors.

True

8. Functions are organised code segments which cannot be reused to implement single or related operations.

False

9. Python tuples are enclosed in parentheses (()) and the internal elements are separated by semicolons (;)

False

10. Matrix A and Matrix B can be added only if they have the same number of rows and columns.

True

11. The determinant of a $N \times N$ matrix is scalar.

True

12. Deep learning is a class of machine learning algorithms that uses neural networks to extract features from the raw input.

True

Name : _____

13. Huawei believes that the value of AI is reflected in addressing enterprise business problems. Regarding the development of AI, guided by industry solutions, Huawei focuses on basic technologies and the enabling platform, and works with industry leaders to develop industry solutions.

True

14. The Python author designed the restrictive syntax to filter out bad programming habit (like failing to add an indentation on the right or the next line of the if statement)

True

15. Convolutional neural networks (CNNs) can extract features from images automatically.

True

16. Continue statements in Python are used to skip the rest of the statements in the body of the current iteration and jump to the beginning of the loop for next iteration.

True

17. Lists in Python can be delimited by Square brackets ([]), with the first element index at 0.

True

18. `time.time()` function can measure the number of seconds since the epoch as a floating point value.

True

19. Principal Component Analysis (PCA) is a statistical method. It uses orthogonal transformation to convert a set of possibly correlated variables into a set of values of linear variables called principal components.

False

20. Recurrent neural networks are more suitable for image recognition problems.

False

Name : _____

21. Which of the following can be the input of a fully connected layer?

- A. All of the above.
- B. A fixed length vector.
- C. A variable-length vector.
- D. A 2-D matrix.

Answer B

22. If the input of a softmax function is [1,2,4,2,1], which of the following options may be the output?

- A. [0.04, 0.20, 0.75, 0.20, 0.04]
- B. [3]
- C. [0.04, 0.10, 0.72, 0.10, 0.04]
- D. [10]

Answer C

23. Decision tree is one of the most popular machine learning algorithms and can be used for both classification and regression problems. How do we choose the best attribute in ID3?

- A. The attribute with the highest Gini index
- B. The attribute with the lowest Gini index
- C. The attribute with the highest Information gain
- D. The attribute with the lowest information gain

Answer C

Name : _____

24. Information theory is a branch of applied mathematics. It studies the quantification of information in a signal. Who was the first to put forward the concept of information entropy?

- A. Claude Elwood Shannon
- B. Marvin Lee Minsky
- C. Herbert A. Simon
- D. Alan Mathison Turing

Answer A

25. Which is not a optimization method in deep learning?

- A. Stochastic gradient decent
- B. Backpropagation algorithm
- C. Principal component analysis
- D. Momentum

Answer C

26. Which of the following Python identifiers is invalid.

- A. var_1
- B. _var1
- C. Var_1
- D. 1var

Answer: D

27. What is the purpose of adding a pooling layer when creating a neural network?

- A. Functioning as a classifier.
- B. Preventing overfitting.

Name : _____

- C. Extracting image feature.
- D. Reduce the size of feature maps.

Answer: D

28. Which of the following formats can define a dictionary in Python?

- A. {"John": Paris, "John":London}
- B. {"Jane":London; "John":London}
- C. {"Jane":London, "John":London}
- D. {"John" : Paris; "John" :London}

Answer C

30. Which of the following type of RNN can be used for machine translation?

- A. One to one
- B. Many to one
- C. One to many
- D. Many to many

Answer: A

31. Which of the following activation functions cannot effectively solve the vanishing gradient problem?

- A. Tanh
- B. ReLU
- C. SoftPlus
- D. Leaky ReLU

Answer: C

Name : _____

32. Which of the following statements about normal distribution is not true?

- A. The larger the standard deviation, the flatter the normal distribution curve.
- B. Normal distribution takes its maximum value at the mean.
- C. The larger the standard deviation, the steeper the normal distribution curve.
- D. A large number of random variables are distributed normally or approximately normally in natural and social phenomena.

Answer: B

32. Which is not an automatic hyperparameter search algorithm?

- A. Model-based hyperparameter optimization.
- B. Stochastic gradient descent
- C. Grid search
- D. Random search

Answer: B

33. Which of the following about matrix is not true?

- A. The transposed matrix of a symmetric matrix A is A .
- B. Multiplying by the identity matrix does not change anything.
- C. The diagonal matrix does not have an inverse.
- D. The transposed matrix of an orthogonal matrix is its inverse.

Answer: C

34. Which of the following is not a supervised learning algorithm?

- A. Support vector machine (SVM)
- B. Principle component analysis (PCA)
- C. Logistic regression
- D. Decision tree

Name : _____

Answer: B

35. Which of the following can void overfitting during model training?

- A. Using the momentum optimizer
- B. Adding the number of convolutional layers
- C. Activation functions
- D. Early stopping

Answer: D

36. Which of the following is unchangable in Python?

- A. Turple
- B. Set
- C. Array
- D. List

Answer: A

37. Which of the following is not a deep learning develepment framework?

- A. CNTK
- B. MXNet
- C. Keras
- D. BAFA

Answer: D

38. Which of the following is not a Python re regular expression module function?

- A. Match
- B. Update
- C. Compile

Name : _____

D. Search

Answer: B

39. Which of the following methods cannot get data into the TensorFlow program?

A. Reading from files

B. Feeding

C. Editing a file format reader

D. Preloaded data

Answer: C

40. Which of the following is not a TensorFlow library Module?

A. tf.nn

B. tf.contrib

C. tf.boost

D. tf.layers

Answer: C

41. With regard to the relationship between frequency and probability, which of the following statements is true.

A. Frequency stabilizes as a value close to probability as the number of repeated experiments increases

B. Frequency is equal to probability

C. Frequency is greater than probability

D. Frequency is smaller than probability

Answer: A

42. Which of the following statement about convolutional neural networks(CNNs) is not true?

A. Max pooling is the most commonly used pooling operation.

Name : _____

- B. A pooling layer contains a set of filters whose parameters need to be learned
- C. A convolutional layer contains a set of filters whose parameters need to be learned
- D. Pooling layers in CNN are implemented to reduce the size of feature maps

Answer: B

43. Which of the following is used as the gate function in LSTM?

- A. ReLU function
- B. Tanh function
- C. Sigmoid function
- D. Softplus function

Answer: C

44. Which of the following can be used to solve regression problems?

- A. Linear regression
- B. K-means
- C. Logistic regression
- D. PCA

Answer: A

45. For object detection problem, which of the following neural network model should be considered first?

- A. CNN
- B. RNN
- C. GAN

Name : _____

D. GNN

Answer: A

46. Which of the following school does reinforcement learning belong to?

A. None of the above

B. Connectionism

C. Symbolism

D. Behaviourism

Answer: D

47. Which of the following is the computational model of TensorFlow?

A. Session

B. Variable

C. Tensor

D. Computational graph

48. Which of the following is not a TensorFlow session?

A. Multiple POST queries

B. Explicitly invoke the session to close a function

C. Use python context manager

D. Explicitly invoke the session to generate a function

Answer: A

49. Which of the following statement about the fetch operation in TensorFlow is true?

A. Only when the graph execution is triggered, you can fetch the tensor values you want

B. Generally, the fetch operation is used together with `tf.placeholder()`.

C. Only a single tensor node can be fetched once

Name : _____

D. You can fetch the tensor values you want once the graph is created.

Answer: A

50. When was AI first proposed?

A. 1960

B. 1956

C. 1916

D. 1946

Answer: B

51. Which of the following cloud services are provided by visual recognition of Huawei Cloud EI?

A. Image recognition

B. Content moderation

C. Optical character recognition (OCR)

D. Image processing

Answer - ABCD

52. What are the commonly used activation functions?

A. relu

B. danish

C. sigmoid

D. tanh

Answer ACD

53. According to constraints, what are the types of optimization problems?

A. Equality constraint

B. No constraint

C. Inequality constrain

D. Semi-constraint

Answer: ABC

Name : _____

54. Which of the following statements about python index are true?

- A. Default index from right to left is numbered from 0
- B. Default index from Right to Left is numbered from -1
- C. Default index from Left to Right is numbered from 1
- D. Default index from Left to Right is numbered from 0

Answer: B, C

55. Which of the following statements about Python is true

- A. Python is often nicknamed "Glue Language" because it can easily glue together various modules developed in other languages
- B. Python is a kind of free and open-sources software.
- C. The Python syntax is simple and clear, characterized by using black characters for indentation.
- D. It was created by Guido van Rossum, the Dutchman in 1989, and it was first released in 1991

Answer: A,B,C,D

56. What are the research areas in AI?

- A. Computer vision
- B. Speech recognition
- C. Natural Language Processing (NLP)
- D. Machine Learning

Answer: A,B,C,D

57. Python is a object-oriented programming language. Which of the following are Python objects?

- A. Character string
- B. Number
- C. dModule
- D. Function

Name : _____

58. Which of the neural network architectures can implement weight sharing?

- A. All of the above
- B. Recurrent neural networks (RNN)
- C. Fully connected neural network
- D. Convolutional neural network (CNN)

Answer: B,D

59. What are the implementation modes of TensorFlow?

- A. Reverse mode
- B. Distributed mode
- C. Forwarding mode
- D. Single-Node mode

Answer: B,D

60. Suppose X and Y are random variables and C is a constant, which of the following variance properties is not true?

- A. $D(C) = 0$
- B. $D(XY) = D(X)D(Y)$
- C. $D(CD) = C \cdot C \cdot D(X)$
- D. $D(X+Y) = D(X) + D(Y)$

6. The determinant of array A is a scalar.

True

False

Dwai - False

9. A model constructed by a machine learning algorithm is a real distribution function of data.

True

Name : _____

False

Answer: True

10、 Spam classification is one of the typical applications of Bayesian theorem.

True

False

Answer: True

11、

TensorFlow is an open-source framework of Google, which supports deep learning algorithms.

True

False

12、

Compared with L2 regularization, L1 regularization can generate a sparser model.

True

False

TRUE

13、

Lists in Python can be delimited by square brackets, with the first element at index 0.

True

Name : _____

False

14、

Python is a dynamically compiled cross-platform language that features fast execution.

True

False

30、 import numpy as np

a = np.arange(12)

b = np.arange(8)

Which of the following can obtain the result after running the program?

A. a.reshape(4,3)*b.reshape(4,2)

B. np.matmul(a.reshape(-1,8)*b)

C. a.reshape(3,4)+b.reshape(4,2)

D. np.matmul(a.reshape(3,4),b.reshape(4,2))

32、

import tensorflow as tf

w1 = tf.Variable(tf.random_normal([2,3],mean=1.0, stddev=1.0))

Name : _____

```
w2 = tf.Variable(tf.random_normal([3,1],mean=1.0, stddev=1.0))
```

```
x = tf.constant([[0.7, 0.9]])
```

```
init = tf.global_variables_initializer()
```

Which of the following can be used to obtain the value of y ($x*w1*w2$) in TensorFlow1.0?

A. `a = tf.matmul(x ,w1)`

```
    y = tf.matmul(a, w2)
```

```
    print(y.eval())
```

B. `with tf.Session() as sess:`

```
    a = tf.matmul(x ,w1)
```

```
    y = tf.matmul(a, w2)
```

```
    print(y.eval())
```

C. `with tf.Session() as sess:`

```
    sess.run(init)
```

```
    a = tf.matmul(x ,w1)
```

```
    y = tf.matmul(a, w2)
```

```
    print(y.eval())
```

D. `with tf.Session() as sess:`

```
    sess.run(x*w1*w2)
```

Multiple choice single answer 37/60Question

Arrow keys available to switch questions

Name : _____

37、

Which of the following statements about the TensorFlow module is incorrect?

- A. `tf.layers`: Encapsulates variables and operations on the variables.
- B. `tf.contrib`: Provides functions for calculating streaming indicators.
- C. `tf.estimator`: High level tools for working with models.
- D. `tf.nn`: Encapsulates data processing functions.

39、

Which of the following statements about probability is correct?

- A. A distribution function describes the value rules of random variables.
- B. The event occurrence probability estimated when the event causes have been known is usually called conditional probability or prior probability.
- C. The probability is also called the frequency.
- D. The distribution law can only describe the value rule of continuous random variables.

Multiple choice single answer 42/60Question

Arrow keys available to switch questions

Name : _____

42、

Which of the following statements about the recurrent neural network is incorrect?

- A. The context state can be saved.
- B. LSTM is also one type of recurrent neural network.
- C. Compared with RNN, LSTM does not have the problem of gradient disappearance.
- D. RNN is widely used in sequence-related scenarios.

1. Matrix A and Matrix B can only be added if they have equal number of rows and columns

True, False

2. Operations and computation graphs of Tensorflow can run without session

True, False

3. In Tensorflow, a graph contains nodes and edges. The former represent data operations and the later represent data and its dependency relationship.

True, False

4. Huawei believes that the value of AI is reflected in addressing enterprise business problems. Regarding the development of AI, guided by industry solutions, Huawei focuses on basic technologies and the enabling platform and works with industry leaders to develop industry solutions

True, False

5. Deep learning is a class of machine learning algorithms that uses neural networks to extract features from the raw input.

True, False

6. `time.time()` function can measure the number of seconds since the epoch as a floating point value.

True, False

7. Convolutional neural networks(CNN) can extract features from images automatically

True, False

8. The determinant of a $N \times N$ matrix is a scalar.

Name : _____

True, False

9. The print function in Python 3 is enclosed in parenthesis (()).

True, False

10. Deep learning algorithms can be both supervised and unsupervised

True, False

11. The perceptron is equivalent to a classifier and it could perform multi-task classification.

True, False

12. Continue statements in Python are used to skip the rest of the statements in the body of loop for the current iteration and jump to the beginning of the loop for next iteration

True, False

13. Lists in Python can be delimited by square brackets ([]), with the first element at index 0.

True, False

14. Principal Component Analysis(PCA) is a statistical method. It uses orthogonal transformation to convert a set of possibly correlated variables into a set of values of linear variables called principal components.

True, False

15. Python tuples are enclosed in parenthesis (()) and the internal elements are separated by semicolons(,)

True, False

16. The Python author designed the restrictive syntax to filter out bad programming (like failing to add an indentation on the right of the next line of the if statement)

True, False

17. Data in Tensorflow is represented by tensors

Name : _____

True, False

18. Functions are organized code segments which cannot be reused to implement single or related operations.

True, False

19. Recurrent neural networks are more suitable for image recognition problems.

True, False

20. Matrix A can be decomposed into a product of three matrices using singular value decomposition

True, False

21. Which of the following methods CANNOT get data into Tensorflow program?

Editing a file format reader

22. Data in Tensorflow is represented by tensors

True, False

23. Which is not an automatic hyperparameter search algorithm?

Grid search

Random search

Stochastic gradient descent?

Model-based hyperparameter optimization

24. Which of the following is not a Python re regular expression module function?

Match

Compile

Update

search

25. Which of the following can be the input of a fully connected layer?

A 2-D matrix

A variable-length vector

All of the above

A fixed-length vector

26. Which is not an automatic hyperparameter search algorithm?

Frequency is smaller than probability

Frequency stabilizes a value close to probability as the number of repeated experiments increases

Frequency is equal to probability

Frequency is greater than probability.

27. Which of the following is not a tensorflow library module?

Tf.contrib

Tf.boost

Tf.nn

Tf.layers

28. When was AI first produced?

1956

Name : _____

29. Which of the following activation functions CANNOT effectively solve the vanishing gradient problem?

Leaky ReLU

Name : _____

Tanh

ReLU

SoftPlus

30. In the input of a softmax function is [1,2,4,2,1], which of the following option may be the output?

[0.04, 0.20, 0.75, 0.20, 0.04]

[0.04, 0.10, 0.72, 0.10, 0.04]

[10]

[3]

31. Which is NOT an optimization method in deep learning?

Stochastic gradient descent

Principle component analysis (PCA)

Backpropagation

Momentum

32. Which of the following Python identifier is invalid?

1var

Var_1

33. Which of the following is the computational model of tensorflow?

Session

Tensor

Computational graph

Variable

34. Which of the following format can define a dictionary in Python?

{“Jane”:London, {“John”:London}}

35. Which of the following statement about the fetch operation in tensorflow is true?

Only a single tensor node can be fetched once

Generally, the fetch operation is used together with tf.placeholder().

Only when the graph execution is triggered, you can fetch the tensor values you want

You can fetch the tensor values you want once the graph is created.

36. Which of the following is NOT a deep learning development framework?

CNTK

BAFA

MXNet

Keras

37. Decision tree is one of the most popular machine learning algorithms and can be used for both classification and regression problems. How do we choose the best attribute in ID3

The attribute with the highest information gain

The attribute with the lowest information gain

The attribute with the lowest Gini index

The attribute with the highest Gini index

38. Which of the following is NOT a supervised learning algorithm?

Principle component analysis (pca)

Logistic regression

Name : _____

Support vector machine (SVM)

Name : _____

Decision tree

39. What is the purpose of adding a pooling layer when creating a neural network?

Reduce the size of feature maps

Extracting image feature

Preventing overfitting

Function as a classifier

40. Which of the following about normal distribution is not true?

The larger the standard deviation, the steeper the normal distribution curve.

41. Which of the following is used as the gate function in LSTM?

Relu function

Softplus function?

Sigmoid function

Tanh function

42. Which of the following about matrix is NOT true?

The diagonal matrix does not have an inverse

43. Which of the following can be used to solve regression problems?

Linear regression

44. Which of the following can avoid overfitting during model training?

Early stoping

45. Information theory is a branch of applied mathematics

Claude Elwood Shannon

46. Which of the following statement about convolutional neural networks is NOT true?

A pooling layer contains a set of filters whose parameters need to be learned

47. Which of the following is NOT a tensorflow session?

Multiple POST queries

48. Which of the following school does reinforcement learning belong to?

Behaviorism

Connectionism

None of the above

Symbolism

49. Which of the following type of RNN can be used for machine translation?

Many to one

Many to many

One to many

One to one

50. For object detection problem, which of the following neural network model should be considered first?

CNN

51. Which of the following is unchangeable in Python ?

Name : _____

Tuple

52. What are the implementation modes of tensorflow?

Distributed mode

Single node mode

Name : _____

53. What are the research areas in AI?

Natural language processing (NLP)
Computer vision
Speech recognition
Machine learning

54. Which of the following statements about Python index are true?

Default index from left to right is numbered from 0
Default index from right to left is numbered from -1

55. Which of the following cloud services are provided by visual recognition of Huawei Cloud EI?

Content moderation
Image processing
Optical character recognition(OCR)
Image recognition

56. What are the commonly used activation functions?

relu
tanh
sigmoid

57. According to constraints, what are the types of the optimization problem?

Equality constraint
No constraint
Inequality constraint

58. Suppose X and Y are random variables and C is a constant, which of the following about the variance properties is not true?

$D(XY) = D(X)D(Y)$

59. Which of the following cloud services are provided by visual recognition of Huawei Cloud EI?

Character string
Number
Function
Module

60. Which of the neural network architectures can implement weight sharing?

Convolutional neural network(CNN)
Recurrent neural network(RNN)

61. Which of the following statements about Python are true?

The Python syntax is simple and clear, characterized by using blank characters for indentation
Python is often nicknamed glue language because it can easily glue together various modules developed in other languages
It was created by Guido van Rossum, the Dutchman, in 1989, and it was first released in 1991
Python is a kind of free and open-source software.

Name : _____

1. (Single-Answer Question) When was artificial intelligence (AI) first proposed? ☐ A. 1946 ☐ B. 1960 ☐ C. 1916 ☐ D. 1956 2. (Multiple-Answer Question) What are the elements of AI? ☐ A. Algorithms ☐ B. Computing performance ☐ C. Data ☐ D. Scenarios 3. (Multiple-Answer Question) Which of the following statements about Python are true? ☐ A. It was created by Guido van Rossum, the Dutchman, in 1989, and it was first released in 1991. ☐ B. Python is a kind of free software and its open-source code complies with the GPL (GNU General Public License) Protocol. ☐ C. The Python syntax is simple and clear, characterized by using blank characters for indentation. ☐ D. Python is often nicknamed "glue language" because it can easily glue together various modules developed in other languages. 4. (True or False) Multiple Python statements can be written in one line with commas (,) between the statements. ☐ A. TRUE ☐ B. FALSE 5. (Single-Answer Question) Which of the following is not a Python operator? ☐ A. Arithmetic operator ☐ B. Inference operator ☐ C. Logical operator ☐ D. Comparison operator 6. (Single-Answer Question) Which of the following statements about Python regular expression is not true? ☐ A. re regular expression module was added in Python 1.5. ☐ B. re module grants all regular expression functions to Python. ☐ C. re regular expression is a powerful tool to process strings. ☐ D. re regular expression can only process character string data, not numerical data.

Name : _____

7. (True or False) The trace operation returns the sum of the diagonal elements of the matrix. So the trace of matrix A is the same as that of its transposed matrix. () A. TRUE B. FALSE 8. (Single-Answer Question) Which of the following is not the main idea of linear algebra? () A. Probability theory B. Linear transformation C. Matrix theory D. Vector space 9. (Single-Answer Question) matrix A is a 3×2 matrix, B is a 2×3 matrix, and C is a 3×3 matrix. Which of the following operations is valid? () A. AC B. BC C. A+B D. AB-BC 10. (True or False) Principal Component Analysis (PCA) is a statistical method. It uses orthogonal transformation to convert a set of possibly correlated variables into a set of values of linearly dependent variables called principal components. () A. TRUE B. FALSE 11. (True or False) Correlation coefficient, also called the linear correlation coefficient, measures a linear relationship between two variables. Its value is a real number greater than 0. () A. TRUE B. FALSE 12. (Single-Answer Question) Suppose X and Y are random variables and C is a constant, which of the following about mathematical expectation properties is NOT true? () A. $E(C)=C$ B. $E(X+Y)=E(X)+E(Y)$ C. $E(CX)=CE(X)$ D. $E(XY)=E(X)E(Y)$

Name : _____

13. (Multiple-Answer Question) What are the characteristics of TensorFlow? ☐ A. Open source B. Convenient C. Well-developed D. Flexible 14. (True or False) Operations and computation graph of TensorFlow do NOT necessarily run in the Session. ☐ A. TRUE B. FALSE 15. (Single-Answer Question) Which of the following is not a Deep Learning development framework? ☐ A. CNTK B. Keras C. BAFA D. MXNet 16. (True or False) GAN is a Deep Learning model. Recently, it is one of the most promising methods for unsupervised learning in complex distribution. ☐ A. TRUE B. FALSE 17. (Single-Answer Question) According to PEP 8, how many spaces should the first line be indented by in Python? ☐ A. 1 B. 2 C. 4 D. 8 18. (Multiple-Answer Question) Training error causes underfitting by reducing model accuracy. How can the model be a better fit? ☐ A. Increasing data volume B. Creating feature engineering C. Reducing regularization parameters D. Adding features

Name : _____

19. (Single-Answer Question) Which of the following statements about gradient descent is NOT true? ()

A. Stochastic gradient descent is a common form of gradient descent. B. Gradient descent includes stochastic gradient descent and batch gradient descent. C. The gradient descent algorithm is fast and reliable. D. Stochastic gradient descent is one of the commonly used optimization algorithms in Deep Learning.

20. (Single-Answer Question) Which of the following is NOT involved in the three types of services for Huawei Cloud EI to construct enterprise intelligent services? () A. Basic platform services A. General domain services B. Industry services C. Integration services D. Answers: 1. Python tuples are identified by "()" and internal elements are separated by ":".

- True

- False

2. What are the numeric types of Python? (Multiple Choice)

- int (integer type)

- long (long integer type)

- float (floating point type)

- complex (complex number type)

3. There are a lot of data generated during the training of the neural network. What mechanism does TensorFlow use to avoid excessive input data?

- - Client

- - feed

- - placeholder

- - fetch

Name : _____

4. What is the English abbreviation for AI?

- • Automatic Intelligence
- • Artificial Intelligence
- • Automatic Information
- • Artificial Information

5. There are many commercial applications for machine learning services. What are the main business scenarios covered? (Multiple Choice)

- • Financial product recommendation
- • Predictive maintenance
- • Telecom customer retention
- • Retailer grouping

6. What is WRONG description of the Python module?

- • The Python module is a Python file that ends with .py and contains Python object definitions and Python statements.
- • The Python module allows you to logically organize your Python code snippets.
- • Python modules can define functions, classes, and variables, but the module does not contain executable code.
- • Assigning relevant code to a module can make your code better and easier to understand.

7. Which of the following options is not the session mode used by TensorFlow?

- • Explicitly call the session to generate function
- • Explicitly call the session to close function
- • Through the Python context manager
- • Multiple POST queries

8. Principal Component Analysis (PCA) is a statistical method. A set of variables that may be related to each other is transformed into a set of linearly related variables by orthogonal transformation. The converted set of variables is called the principal component.

- • True
- • False

Name : _____

9. The history of robots is not long. In 1959, the United States, Engelberg and Devol made the world's first generation of industrial robots, and the history of robots really began. According to the development process of the robot, it is usually divided into three generations, respectively are: (Multiple Choice)

- • Teaching Reproduction Robot
- • Robot with sensation
- • Robots that will think
- • Intelligent robot

10. As the following, what are the Python language design philosophy? (Multiple Choice)

- • Beautiful
- • Expensive
- • Explicit
- • Simple

11. What are the service categories included in Huawei Cloud EI Enterprise Intelligence? (Multiple Choice)

- • EI visual cognition
- • EI online games
- • EI speech semantics
- • EI industry scene

12. What of the following does belong to convolutional neural network (CNN)? (Multiple Choice)

- • VGGNet
- • ResNet
- • AlexNet
- • GoogleNet

13. What is wrong description of the normal distribution?

- • In natural phenomena and social phenomena, a large number of random variables obey or approximate a normal distribution.
- • The normal distribution takes the maximum value at the mean.
- • The larger the standard deviation, the steeper the normal distribution curve.
- • The larger the standard deviation, the slower the normal distribution curve.

Name : _____

14. In May 1997, the famous “Human-Machine Wars” final computer defeated Kasparov, the world chess king, with a total score of 3.5 to 2.5. Is this computer called?

- • Dark blue
- • Dark green
- • Ponder
- • Blue sky

15. What is WRONG description of backpropagation?

- • The learning process of the backpropagation algorithm consists of a forward propagation process and a back propagation process.
- • The backpropagation algorithm is a learning algorithm suitable for multi-layer neural networks, which is based on the gradient descent method.
- • The backpropagation phase sends training inputs to the network to obtain an stimuli response.
- • The backpropagation algorithm is mainly repeated by two loops (excitation propagation, weight update) until the response of the network to the input reaches the predetermined target range.

16. Which of the following options is NOT a reason for traditional machine learning algorithms to promote the development of deep learning?

- • Dimensional disaster
- • Local invariance and smooth regularization
- • Manifold learning
- • Feature Engineering

17. The commonly used functions for mathematical operations in Python are basically in the math module and the cmath module.

- • True
- • False

18. What does NOT belong to supervised learning?

- • Logistic regression
- • Support vector machine
- • Decision tree
- • Principal component analysis

Name : _____

19. Functions are well-organized, non-reusable code segments used to implement a single, or associated function.

- True
- False

20. What is the subject of artificial intelligence?

- Mathematics and Physiology
- Psychology and Physiology
- Linguistics
- Comprehensive interdisciplinary and marginal disciplines

21. Which of the following conditions is NOT a condition to be satisfied by the n-fold Bernoulli test?

- Each test itself must follow a normal distribution
- There are only two possible outcomes per trial, that is, event A occurs
- The results of each test are independent of each other
- Each test is repeated under the same conditions

22. What is INCORRECT about the following matrix?

- Diagonal matrix does not exist inverse matrix
- Multiplying any matrix with the identity matrix will not change
- The symmetric matrix A is transposed or A itself
- The orthogonal matrix A's transpose is equal to the inverse of A

23. What is WRONG with the introduction of Python module time?

- from time
- import time
- from time import*
- import time as t

24. In Tensorflow, data is expressed in the form of tensors and computational graphs

- True
- False

Name : _____

25. Which of the following is NOT a specific technology of artificial intelligence?

- Riemannian geometry
- Semantic understanding
- Knowledge graph
- Machine translation

26. Tensorflow is Google's first generation of proprietary machine learning system.

- True
- False

27. Which of the following options is NOT a tensor attribute?

- node
- type
- shape
- name(identification)

28. The following statements about overflow and underflow in numerical calculations are correct?

- Underflow occurs when the numbers close to zero are rounded to zero
- Overflow occurs when a large number of levels are approximate to positive infinity or negative
- Underflow occurs when a large number of levels are approximate to positive infinity or negative
- An overflow occurs when the numbers close to zero are rounded to zero

29. The neural network training process will generate a lot of data. What mechanism does Tensorflow use to avoid the problem of too much input data?

- placeholder
- fetch
- feed
- client

30. What are the advantages of Python?

- free
- simple
- high-level language
- rich library

Name : _____

31. What are the Python language data types?

- • List
- • string
- • tuple
- • number
- • dictionary

32. What is NOT the difference between Python 2 and Python 3?

- • import
- • print
- • unicode
- • xrange

33. Which of the following is NOT a module in the Tensorflow library?

- • tf.boost
- • tf.nn
- • tf.layers
- • tf.contrib

34. Principal Component Analysis (PCA) is a statistical method. Through orthogonal transformation, a set of variables that may have correlation is converted into a set of linearly related variables. The converted set of variables is called the principal component.

- • False
- • True

35. What operating systems does Python support?

- • linux
- • DOS
- • Windows
- • MacOS

36. What are the implementation modes of Tensorflow?

- • Reverse mode
- • Forward mode
- • Stand – alone mode
- • Distributed mode

Name : _____

37. Print in Python 3 must be used with ()

- True
- False

38. What the convolutional neural networks(CNN)?

- VGGNet
- ResNet
- AlexNet
- GoogleNet

39. Which of the following environments does NOT support the installation of Tensorflow?

- Openstack
- Linux
- MacOS
- Docker

40. The Python script execution model includes interactive mode and script mode

- True
- False
- 0
- Null

41. The components of the Python identifier include: numbers, letters, and underscores

- True
- False

42. What is INCORRECT about the following description of the distribution function, distribution law, and density function of random variables?

- Discrete random variables have no distribution function
- Distribution function describes the law of random variables
- The distribution law can only describe the law of the value of discrete random variables
- The distribution law can only describe the law of the value of discrete random variables

43. The autoencoder is an unsupervised learning algorithm

- True
- False

Name : _____

44. Python tuples are identified by “()” and internal elements are separated by “:”.

- True
- False

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- • Feature Engineering

Name : _____

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- ☐ True
- ☐ False

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- ☐ Support vector machine
- ☐ Decision tree
- ☐ Principal component analysis

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